WEAK ETALENESS IS CHECKED ON THE LOCAL RINGS

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 18901-18916, immediately following the statement at 18891-18899.

Editorial note. One cross-chapter pointer (Algebra) is recorded in assembly.external_references.

Editorial note. The displayed chain of canonical maps is retained verbatim; it is the technical heart of the proof.

Allowed standard prerequisites (editorial).

- Flatness versus localisation of a ring map. Algebra, lemma algebra-lemma-flat-localization part (2), cited at proof line 18915. Provides the equivalence between flatness of a composition and flatness of one factor, which is the algebraic content of the proof.
- Weak etaleness of a morphism and of a ring map. the definitions used in conditions (1) and (2) of the statement. Fix what the two conditions mean, in particular that both force flatness.
- Canonical maps of local rings of the diagonal. the displayed chain at proof lines 18905-18911. Turns the two conditions into flatness statements about the first and the composite arrow of that chain.

Source reference. Original reference 00HT: focused-seeds/geometry/S9-morph-check-weakly-etale-stalks/ref-tags.txt, source lines 4–4. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $f : X \to Y$ be a morphism of schemes. The following are equivalent*

(1) *$X \to Y$ is weakly étale, and*
(2) *for every $x \in X$ the ring map $\mathcal{O}_{Y,f(x)} \to \mathcal{O}_{X,x}$ is weakly étale.*

Proof. Observe that under both assumptions (1) and (2) the morphism f is flat. Thus we may assume f is flat. Let $x \in X$ with image $y = f(x)$ in Y. There are canonical maps of rings

$$\mathcal{O}_{X,x} \otimes_{\mathcal{O}_{Y,y}} \mathcal{O}_{X,x} \longrightarrow \mathcal{O}_{X \times_Y X, \Delta_{X/Y}(x)} \longrightarrow \mathcal{O}_{X,x}$$

where the first map is a localization (hence flat) and the second map is a surjection. Condition (1) means that the second arrow is flat for all x. Condition (2) is that the composition is flat for all x. Thus the equivalence by Algebra, Lemma 00HT part (2). □

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.